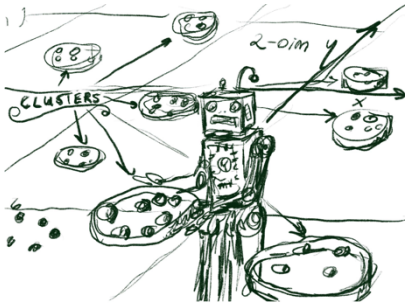
LEIC-T 2024/2025
Aprendizagem - Machine Learning
Homework 4
Deadline 22/10/2025 23:00
Submit on Fenix as pdf



I) (7 pts) Clustering

Given the data

$\mathbf{x}_1 = \begin{pmatrix}0\\0\end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix}2\\0\end{pmatrix}, \mathbf{x}_3 = \begin{pmatrix}0\\3\end{pmatrix},$

$\pi_1 = 0.6, \pi_2 = 0.4$

$C_1\left(\mathbf{u}_1 = \begin{pmatrix}1\\1\end{pmatrix}, \mathbf{\Sigma}_1 = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\right),\ C_2\left(\mathbf{u}_2 = \begin{pmatrix}-1\\-1\end{pmatrix}, \mathbf{\Sigma}_2 = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\right).$

i) (6 pts)
Perform one iteration of EM clustering algorithm step by step and determine the new parameters. Indicate all the calculations step by step. (To make the calculation easier for each step you can use a computer, however you should be able to do it by hand)
ii) (1 pts)
Performing a hard assignment of observations to clusters identify the silhouette of the larger cluster

II Software Experiments (3pts)

Download the jupyter notebook HM4_25_CL.ipynb.

a) (1 pts)
Load the build in data set "breast_cancer". Perform k-Means and EM clustering with 2 clusters and indicate the silhouette as defined in the notebook. Which clustering, k_means or EM-Clustering is better?
b) (1 pts)
Load the build in data set "breast_cancer". Perform k-Means and EM clustering with 5 clusters and indicate the silhouette as defined in the notebook. Why are the results worse (better) than when we clustered with 2 clusters in a)? (one sentence)
c) (0.5 pts)
Perform PCA with two components on the data set "breast_cancer".
Plot the scatter plot. Can the two classes be separated? (one sentence)
d) (0.5 pts)
Perform k-means and EM clustering with 2 clusters on the PCA generated data set in question c) and indicate the silhouette as defined in the notebook. Plot the scatter plot of the clustering as in the notebook. Which clustering, k_means or EM-Clustering is better? Why are the silhouette values different to the values in a)? (one/two sentence/s)